

Stephen O'Neil

Full Stack Web Application Developer

11+ years frontend | 8+ years backend & DevSecOps

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- A senior developer & technical advisor, modernizing Canada's public health data platforms & infrastructure
- Previously lead developer on GC InfoBase (canada.ca/gcinfobase, github.com/TBS-EACPD/infobase), a React SPA client backed by a serverless GraphQL API

Resume

Senior Developer & Technical Advisor (01/2023 - Present)

Digital Transformation Branch, Health Canada 07/2024 - Present

- Architected and lead development on the [Interoperable Immunization Data Initiative](#) proof of concept, demonstrating interoperable immunization data exchange using FHIR-based APIs, federated aggregation, and simulated cross-jurisdictional record transfers between provincial and federal stakeholders

Data, Surveillance and Foresight Branch, Public Health Agency of Canada 06/2023 - 07/2024

- Contributed logging utilities to the [branch's internal Django library](#), enabling standardized, enriched, and security policy compliant logs in both Google Cloud & Azure environments
- Lead infrastructure and security architecture design and implementation for the branch's first Kubernetes-based Django application deployment
 - Coordinated with project managers, application developers, and IT security to collect and track business and technical requirements; drafted the resulting architecture
 - Achieved a production-ready deployment and provided lessons-learned for future kubernetes use within the organization

Science and Parliamentary Infrastructure Branch, Public Services and Procurement Canada 01/2023 - 06/2023

- Advised on technical aspects of a branch-wide data policy refresh
- Triaged an inherited portfolio of ~20 projects in various stages from prototype to pilot
 - Identified and decommissioned dead projects, directly reducing cloud bills, maintenance burden, and potential risks & attack surfaces
 - Matured the remaining applications and infrastructure by eliminating manual deploy steps, implementing automated rollback capabilities, hardening configurations, etc.

From Developer to Lead Developer (09/2017 - 01/2023)

Expenditure Management Sector, Treasury Board of Canada Secretariat

- Lead the [GC InfoBase](#) project beginning in Summer 2018; held responsibility for the project's technical direction, overall design, and for the team's developer talent
 - Developed and maintained the project's large React frontend
 - Built backend microservices to enable new features in the core application
 - Architected and administered infrastructure and operations for the project
 - E.g. transitioned hosting from tenancy in an external team's servers to cloud platforms & an in-house DevOps culture, taking deploy times from days to minutes
 - Developed tooling to multiply team productivity and reinforce best practices
 - E.g. implemented automated end-to-end WCAG 2.1 accessibility testing for CI
 - Championed code standards, review practices, testing, CI/CD, and working in the open
 - E.g. sold stakeholders on open sourcing the project; was awarded in recognition of the direct positive impacts one year later
 - Hired, supervised, and mentored junior developers and CO-OP students
- Built the data collection tool used for the centralized tracking of COVID-19 related expenditures across all government departments
 - Quickly learnt Python, Django, an existing internal platform's code & environment, and the business requirements; met the ambitious 3 month deadline provided for launch
 - Contributed lasting improvements to the existing platform's testing practices

Student & Part-time Developer (01/2015 - 09/2017)

Office of the Chief Human Resources Officer, Treasury Board of Canada Secretariat

- Self-taught JS, HTML, CSS, and Git; developed the early HR data portions of GC InfoBase

Education

Honours B.Sc. Physics-Mathematics, uOttawa
2017, Cum Laude

Certifications

Google Cloud Certified - Professional Cloud Architect
(tinyurl.com/oneil-gc-cert)

Languages

- TypeScript & JavaScript
- HTML ➤ CSS & Sass
- Python ➤ SQL
- sh / Bash / POSIX shells

Libraries &

Frameworks

- React ➤ GraphQL
- Lodash ➤ Jest
- React Testing Library
- Astro ➤ Axe-core
- NextJS ➤ Django
- OpenTelemetry

Tools & Software

- Git ➤ VSCode
- Linux ➤ Docker
- PostgreSQL ➤ MongoDB
- Redis ➤ Kubernetes

Platforms

- Google Cloud Platform
- DigitalOcean ➤ CircleCI
- GitHub Actions

Other Skills & Competencies

- Architecture ➤ Automation
- Application security
- Accessibility ➤ Mentorship
- Strategic leadership